CENTOS 7 SQUID PROXY SERVER SETUP

1. Introduction

- This document provides a complete step by step guide to installing, configuring, and testing a Squid Proxy Server on CentOS 7. The server is configured to allow HTTP/HTTPS traffic from a local network and tested using a client machine.

2. Prerequisites

- CentOS 7 installed (Physical / Virtual Machine)
- Root or sudo access
- Internet connectivity for package installation
- Basic knowledge of Linux terminal commands

3. Step I: Log in as Root access

- If you are logged in as a normal user (e.g. sam), switch to root:
  - su -
- Enter the root password when prompted.

Step II: Verify Internet Connectivity

- Before installing packages, ensure the server has internet access:
  - ping -c 4 8.8.8.8
- If DNS fails, configure DNS in:
  - /etc/resolv.conf:
  - vi /etc/resolv.conf713
- Add:
  - nameserver 8.8.8.8
  - nameserver 1.1.1.1
- Save and exit
  - (Esc, :wq)
- Enter.

Step III: Fix CentOS 7 Repositories (Vault)

- CentOS 7 is End-of-Life, so default mirrors are down. Configure the Vault repository:
  - mkdir /etc/yum.repos.d/backup
    mv /etc/yum.repos.d/CentOS-* /etc/yum.repos.d/backup/
  - cat > /etc/yum.repos.d/CentOS-Vault.repo << 'EOF'
    [base]
    name=CentOS-7 - Base
    baseurl=http://vault.centos.org/7.9.2009/os/$basearch/
    gpgcheck=1
    gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7

```
[updates]
name=CentOS-7 - Updates
baseurl=http://vault.centos.org/7.9.2009/updates/$basearch/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7

[extras]
name=CentOS-7 - Extras
baseurl=http://vault.centos.org/7.9.2009/extras/$basearch/
gpgcheck=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-CentOS-7
EOF
```

- Verify:
  - yum repolist all

Step IV: Install Squid

- yum install -y squid
- Check installation:
  - squid -v

Step V: Configure Squid

- Edit the main configuration file:
  - vi /etc/squid/squid.conf
- Modify the following directives (use / to search within vi):
  - # Define local network
    acl localnet src 192.168.1.0/24
  - # Allow local network and localhost
    http_access allow localnet
    http_access allow localhost
  - # Deny all other access
    http_access deny all
  - # Proxy port
    http_port 3128
- Save and exit
  - Esc, :wq
- Enter

Step VI: Start and Enable Squid

- systemctl start squid
- systemctl enable squid
- systemctl status squid
- Ensure it shows Active: active (running).

Step VII: Configure Firewall & SELinux

- Open port 3128 in the firewall:
  - ❖ firewall-cmd --permanent --add-port=3128/tcp
    firewall-cmd --reload
- If SELinux is enforcing:
  - ❖ setsebool -P squid_connect_any on

Step VIII: Test the Proxy Server (Server-Side)

- Test locally:
  - ❖ curl -x http://localhost:3128 http://www.google.com
- If HTML code appears, the proxy works.

Step IX: Create a Client User for Testing

- ❖ useradd client1
- ❖ passwd client1
- Switch to the client user:
  - ❖ su - client1
- Test the proxy:
  - ❖ curl -x http://192.168.1.100:3128 http://www.google.com

Step X: Monitor Live Traffic

- Open a new terminal, switch to root and run:
  - ❖ tail -f /var/log/squid/access.log